Satisfiability implies consistency for natural deduction

The Open Logic Project contributors

openlogic:first-order-logic/natural-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author corollary labelled cor:consistency-soundness (2385-2388); the proof span 2390-2401 is the author proof environment that immediately follows it.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit first-order-logic/natural-deduction, counted once: satisfiability implies consistency. Only the counted corollary and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Inconsistency in natural deduction. A set Gamma is inconsistent when falsity is derivable from undischarged assumptions in Gamma.. It is the assumption the contrapositive starts from (2392-2395).
- Soundness of natural deduction. Everything derivable from Gamma is entailed by Gamma; the unit proves this as its soundness theorem.. It forces every satisfying structure to satisfy the derived falsity (2395-2398).

Source reference. Original theorem thm-soundness: sources/olp-first-order-logic-natural-deduction/pinned-original-source.tex, source lines 2123–2123. The author proof cites the unit's soundness theorem. The load-bearing work of soundness is carried by that separately proved theorem of the same unit; the counted proof retains its own contrapositive argument in full.

Corollary 0.1. *If Γ is satisfiable, then it is consistent.*

Proof. We prove the contrapositive. Suppose that Γ is not consistent. Then $\Gamma \vdash \bot$, i.e., there is !!aderivation of $\bot$ from !!undischarged assumptions in Γ. By thm-soundness, any !!structure $\mathfrak{A}M$that satisfies Γ must satisfy $\bot$. Since $\not\models M$ for every !!structure $\mathfrak{A}M$, no $\mathfrak{A}M$can satisfy Γ, i.e., Γ is not satisfiable. □

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